

Rushiraj Gadhvi

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 🏠 [Github](#) 🎓 [Google Scholar](#)

Technical Skills

Programming Languages: Proficient with Python (PyTorch, gym, transformers, spaCy, NLTK), C++, Dart, SQL
Web/App Frameworks: Flutter, Django, Next.js, Vue.js, Tailwind, FastAPI
Other: Docker, Bash, Git, Linux, GCP, Scientific Writing, Electronics Prototyping (RPi, Arduino, PCB Design)

Experience

RESEARCH COLLABORATOR @HCI Engineering Group, MIT

- Collaborating with [Prof. Stefanie Mueller's](#) group on research project in human-computer interaction and digital fabrication.

Aug 2025 – Present

RESEARCH INTERN @MuRAL Lab, Plaksha University

- Developed energy-efficient navigation system for autonomous surface vehicles as part of the Khalasi Project under [Prof. Sandeep Manjanna](#).
- Built simulation frameworks to evaluate energy-efficient trajectory planning in dynamic aquatic environments.

Jun–Aug 2025

FOUNDER @Housify (Techsurf Solutions)

- Designed and launched a fintech platform to streamline residential society administration and increase transparency.

Jan – Jun 2025

RESEARCH INTERN @HTI Lab, Plaksha University

- Leading research on PosePilot (real-time posture correction system) and AdaptAI (multimodal productivity system) under [Prof. Siddharth](#).
- Published *two papers in top conferences including CHI, 2025*.

Jan 2024 – Feb 2025

CONTRIBUTOR @Google Summer of Code (GSoC) 2024

- Contributed to QuTiP ([project webpage](#)), Improved the visualization tools for quantum circuits in the open-source QuTiP library, widely used in quantum information research. Mentored by: Dr. Boxi Li, [Dr. Neill Lambert](#), [Dr. Alex Pitchford](#)

Jun – Aug 2024

DATA SCIENCE INTERN @BlinkX by JM Financial, Mumbai

- Designed and developed an end-to-end internal tool for processing application feature-based opinions from customer reviews to support strategic decision-making using LLMs, RAG and NLP-based clustering techniques.

Jun – Jul 2024

RESEARCH INTERN @Physical Research Laboratory (PRL), Ahmedabad

- Fine-tuned models for spectral data composition analysis using statistical techniques and developed a Python based spectrum visualization platform that streamlined data exploration and reduced analysis time for large datasets.

Jun – Jul 2023

Education

Plaksha University

B.Tech in Computer Science & Artificial Intelligence (CSAI)

Mohali, India

2021 – 2025

Edunova High School

11th/12th (A-Stream: Math, Physics, Chemistry, Computer)

Ahmedabad, India

2019 – 2021

Achievements

- Selected for **Google Summer of Code 2024 @NUMFOCUS** — One amongst 1,220 selected globally, out of 43k+ applicants
- **AWS AI & ML Scholar Advance 2024** — One amongst 500 selected globally

- **AWS AI & ML Scholar 2023** — One amongst 2,000 selected globally, out of 70,000+ applications
- **UN Millennium Fellowship 2022** — One amongst 3,000 selected globally, out of 31,000+ applications
- **1st rank** amongst 190+ teams at **IIT Bombay E-Cell BnB Event 2022**
- Selected by **Student Startup and Innovation Policy (SSIP) 2020, Gujarat** amongst 200+ projects

Projects

Khalasi [🔗](#) (under review, ICRA'26)

Nov 2024 – present

Energy-Efficient Navigation for Surface Vehicles in Vortical Flow Fields

- Developed a **continuous control RL framework** able to optimize energy consumption in dynamic vortical environments by introducing "flow-surfing" behavior. Achieving **30-50% reduction in energy consumption**.
- Architected a hybrid **Soft Actor-Critic (SAC)** pipeline dealing with partial observability; integrated **Gaussian Process Regression (GPR)** for local flow reconstruction.
- Demonstrated **robust zero-shot generalization by training on variety of custom environments** and deploying on **real-world ocean current data and complex synthetic gyre formations** without fine-tuning.

AdaptAI [🔗](#) (CHI'25)

Nov 2024 – Feb 2025

A Personalized Solution to Sense Your Stress, Fix Your Mess, and Boost Productivity

- AdaptAI is a multimodal AI system that integrates real-time data from egocentric vision, audio, and physiological signals to enhance productivity and well-being by tailoring interventions to individual needs.
- Developed a **tone-adaptive LLM conversational agent that conditions its linguistic style on real-time physiological stress signals**, enabling supportive, context-sensitive dialogue grounded in HRV-based user state modeling.
- Conducted a **15-participant pilot study**, demonstrating **significant gains in task throughput and user satisfaction**.

PosePilot [🔗](#) (IbPRIA'25)

Aug 2023 – Feb 2024

An Edge-AI Solution for Posture Correction in Physical Exercises

- Designed a **sequence-modeling pipeline** using a lightweight **BiLSTM + Attention architecture** to interpret human-joint movement sequences, enabling fast and accurate error detection during physical exercises.
- Developed a **feedback-generation module that transforms model outputs into concise, actionable visual rendering**. Addressed the need for a real-time, personalized posture correction system with applications in sports, rehabilitation, and fitness training.
- Achieved **97.5% pose recognition accuracy** and high confidence in posture correction, **surpassing SOTA** for recognition. **Released a new, high-quality yoga video dataset**.

Publications

Khalasi: Energy-Efficient Navigation for ASVs in Turbulent Vortical Fields

ICRA, 2026

Rushiraj Gadhvi, Sandeep Manjanna

Under Review

[arXiv](#) [🔗](#)

PosePilot: Edge-AI Solution for Posture Correction in Physical Exercises

IbPRIA, 2025

Rushiraj Gadhvi, Priyansh Desai, Siddharth

Accepted

[DOI](#) [🔗](#) — [arXiv](#) [🔗](#)

AdaptAI: A Personalized Solution to Sense Your Stress, Fix Your Mess, and Boost Productivity

CHI, 2025

Accepted

Rushiraj Gadhvi, Soham Petkar, Priyansh Desai, Siddharth

[DOI](#) [🔗](#) — [arXiv](#) [🔗](#)

QuTiP 5: The Quantum Toolbox in Python

Physics Report, 2026

Neill Lambert, Eric Giguère, Paul Menczel, .. Rushiraj Gadhvi .., Franco Nori *et al.*

Accepted

[DOI](#) [🔗](#) — [arXiv](#) [🔗](#)

Certifications

- Nanodegree in AWS Machine Learning Fundamentals ([Certificate](#) [🔗](#))
- Nanodegree in AI Programming with Python ([Certificate](#) [🔗](#))